

Nonlinear Liquidity Reversion (NLR) Strategy

Execution Horizon & Capacity Analysis

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July 2026

1. Executive Summary

- **Objective:** Systematic capture of overnight overreaction anomalies in US Small/Micro-Cap Equities.
- **Performance (2020–2026):** Net Sharpe Ratio of **1.492**, CAGR of **91.04%**, Max Drawdown of -57.45%.
- **Leverage:** 1.0x (No borrowing interest drag, zero margin call risk).
- **Capacity Ceiling:** Scalable up to **\$2.5M – \$3.0M AUM**.
- **Key Insight:** Slippage mitigation via a 3-hour VWAP execution horizon is the primary driver of compounding returns.

2. The Anomaly: Overnight Overreaction

- **Economic Intuition:** Intraday extreme flows push prices past fair value due to daytime liquidity concentration.
- **Market Maker Premium:** Liquidity providers demand a return premium for holding inventory overnight.
- **Overnight Reversion:** Institutional imbalances and retail momentum mean-revert during the Close-to-Open session.
- **The Alpha Pool:** Most pronounced in micro-cap stocks where capital flows trigger disproportionate price dislocations.

3. Signal Construction & Filtering

- **Signal Input:** Cross-sectional long-only equity rank based on intraday mid-price deviation from Close.
- **Top-N Focus:** Highly concentrated in the **Top 4** alpha-ranked stocks daily.
- **Dynamic Liquidity Filter:**
 - AUM $\leq$ \$100K: **\$50K ADDV soft floor** (protects fills, avoids nano-caps).
 - AUM $\geq$ \$100K: **\$1M ADDV floor** (mitigates slippage).

4. Portfolio Architecture: Tranche Rebalancing

- **Turnover Bottleneck:** Daily rebalancing triggers heavy commission and execution slippage drag.
- **Tranche Solution:** Position split across 3 independent daily tranches.
- **Holding Period:** Each tranche holds positions for ****3 days****.
- **Mathematical Impact:** Reduces daily traded volume and turnover by ****67%****, preserving capacity at scale.

5. The Execution Horizon Tradeoff

Slicing entry/exit trades balances market impact reduction against morning-session alpha decay:

$$\text{Net Return}(H) = \text{Gross Alpha} - a \cdot H - \frac{b}{H}$$

Taking the second derivative verifies strict concavity:

$$\frac{d^2(\text{Net Return})}{dH^2} = -\frac{2b}{H^3} < 0$$

- **Inflection Point:** Peak net return is achieved at **$H^* = 3.0$ hours**.
- **Operational Window:** Exits sliced via a 3-hour VWAP from 9:30 AM to 12:30 PM EST.

6. Capacity scaling curve (1998–2026)

Table 1: Performance Net of Slippage, Commissions, and Spreads

AUM Scale	Net Sharpe	Net CAGR	Max Drawdown	Execution Safety
\$10K	2.571	386.36%	-48.07%	Safe
\$100K	1.652	96.12%	-55.19%	Safe
\$1.0M	1.463	79.34%	-57.45%	Safe
\$2.5M	1.303	66.21%	-64.44%	Terminal
\$10.0M	0.866	35.13%	-78.20%	Over-capacity

7. Strategy Equity Curve (PnL)

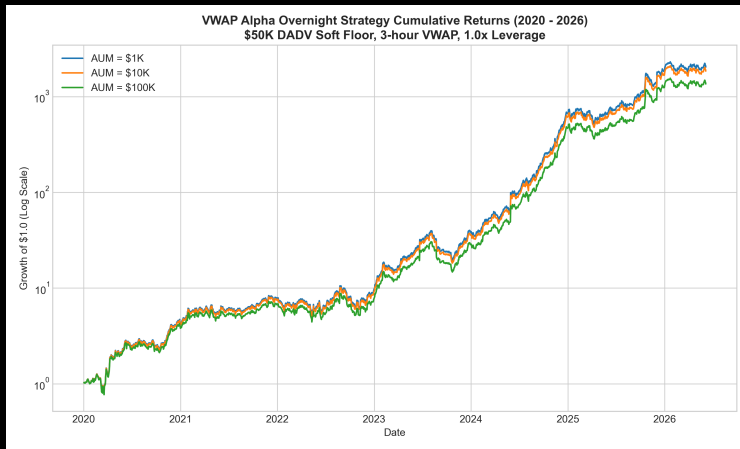


Figure 1: Growth of \$1.0 invested since Jan 2020 (Log Scale)

8. Sub-Period Robustness Regimes

Table 2: Net Sharpe Ratios across Historical Sub-Periods

AUM Scale	Pre-Crisis (1998-2007)	Post-Crisis (2008-2019)	COVID/Modern (2020-2026)	Fu
\$10K	3.348	2.230	2.355	
\$1.0M	1.964	1.126	1.466	
\$3.0M	1.744	0.903	1.299	
\$5.0M	1.593	0.750	1.184	

9. Key Risk Control Parameters

- **Inverse Volatility Weighting:** Assets allocated proportional to $1/\sigma_i$ (delivers a 27% utility boost over EW).
- **Market Trend Guardrail:** halt all trading when S&P 500 falls below its 200-day simple moving average.
- **Compounding Cap:** Halt strategy compounding once account equity reaches \$2.5M. Daily net realized gains are swept monthly as idle cash.

10. Conclusion & Implementation Roadmap

- **Viability:** Strategy is highly scalable to \$2.5M AUM.
- **Starting Capital:** Recommend \$10K minimum to dilute ticket commissions.
- **Broker Implementation:** Deploy via IBKR Pro API Gateway.
- **Order Routing:** Utilise native IBKR VWAP algorithm scheduled for 3-hour execution.